PSAI_003

Open Grant Proposal: Enterprise preprocessor layer for IPFS client (encryption, compression and other enterprise use cases)

Name of Project: Enterprise preprocessor layer for IPFS client (encryption, compression and other enterprise use cases)

Proposal Category: app-dev

Proposer: Andy ZHOU

Do you agree to open source all work you do on behalf of this RFP and dual-license under MIT and APACHE2 licenses?: Yes

Project Description

We plan to make a preprocessor gateway for IPFS client with the aim to fulfil common needs and concerns of enterprise users. So as to attract more enterprise users to embrace IPFS in their infrastructure.

This preprocessor gateway is going to provide ReSTful interfaces over HTTP protocol that is easily accessible by mainstream enterprise applications. Our target audiences will integrate it with the technologies they familiar with.

We plan to carry out our work based on Go-IPFS project, where we add a standalone preprocessing layer before it.

This project will start from the very basic features and extend it in the future projects. The initial features include:

1. Encryption and decryption of files
2. Common open-source encryption algorithms will be supported, also leaving the possibility for future extension
3. Compression and decompression of files
4. Common open-source compression algorithms will be supported, also leaving the possibility for future extension
5. Support of AWS KMS and Aliyun KMS, also leaving possibility to plug in services from the other cloud providers
6. Support local key management
7. Develop a file formant that adds metadata over encrypted/compressed file to facilitate indexing and searching
8. Metadata may include filename, mime type, extension, creation time, key words, description, etc
9. Adopter may choose to fill the metadata or leave it out for anonymity reason

Value

Currently, the cost for a non-blockchain enterprise to adopt Filecoin/IPFS is fairly high, especially the need to understand blockchain architecture, set up FIL wallet, purchase for FIL, set up Filecoin/IPFS node, manage cryptography keys, carry out development projects to fulfil regulation requirements, and etc. All these keep quite a number of potential enterprise users away.

Nevertheless, we foresee that enterprise users will stand a very important role in Filecoin ecology in the future. So we decided to propose this project, with the mission to fulfil the common needs and concerns of enterprise users (encryption, compression, meta data, etc.), letting the enterprise users to embrace IPFS at a marginal cost.

By lowering the “entrance fee” for enterprise users, there will be more and more of them joining IPFS, in the long run, it is for sure some of them will eventually migrate to Filecoin to pursue for higher quality of service. With the strong payment ability the enterprise users have, we believe they will bring

great benefits to Filcoin’s incentive economy.

Deliverables

1. An IPFS preprocessor gateway exposing ReSTful interfaces over HTTP protocol, that allows enterprise applications to interact with.
2. Documentation

Development Roadmap

#	Milestone	Fund	Est. time
1	Technical design in detail and prototyping	US$ 2,000	2 weeks
2	Code: Encryption, decryption and local key management	US$ 6,000	3 weeks
3	Code: KMS integration	US$ 4,000	2 weeks
4	Code: Compression	US$ 2,000	1 week
5	Code: Metadata and file format	US$ 4,000	2 weeks
6	Integration, clean up	US$ 2,000	1 week
7	testing, documentation & publish on Github	US$ 4,000	2 weeks
8	Maintenance for one year	US$ 6,000	1 year

There will be 2 main developers working on all the tasks throughout the project, and we may put in more resources depending on the situation. There will be 1 developer responsible for maintenance.

p/s: The estimated time could look a bit long, because we mainly contribute to this project off work.

Total Budget Requested

US$ 30,000

Maintenance and Upgrade Plans

We will collect feedback and suggestions to keep improving this project.

In case there are resource intensive work, we will consolidate them into a project and apply for a new grant.

In the future, we probably will extend the scope of this project to Filecoin when the time has come.

Team

Team Members

- Andy ZHOU - Lead developer
- Cody HUANG - Developer & project manager
- (May assign more developer from our company depending on the situation)

Team Website

https://www.eidledger.com/

Relevant Experience

Our company, Beijing Gong Yi Lian technologies limited inc. (北京公易联科技有限公司), is a company providing citizen digital identity authentication services that is anonymous and trustable at the same time. Part of our business focuses on enterprise customers (toB), which gain gain us a lot of experiences on integration with enterprise applications and also we know the concerns of our enterprise customers.

Team code repositories

ChatIt - A chat room application that supports 100K simultaneous users on a single server

QuartzCluster - Spring boot Quartz cluster

Additional information

We have integrated IPFS into our infrastructure, nevertheless our work is not open-source-able due to information security policies. Since we find IPFS useful, we decided to give back by starting this project, with the hope that more enterprise users could benefit from IPFS.

Our company is currently leading a project in IEEE standard association, that is P3210 Standard for Blockchain-based Digital Identity System Framework. We see the trend of more and more artists store their NFT masterpieces on Filecoin/IPFS, where they need a way to declare their copyright. With the standard we are working on, the artists will be able to protect their copyright under the identity issued by any regime, thus easily recognized by the law systems. So, we are exploring the possibility to implement such identity standard on Filecoin/IPFS in the future.